AJEENKYA D Y PATIL UNIVERSITY — SCHOOL OF ENGINEERING

BTECH (CTIS) | Sem VII — Research Methodology (Continuous Assignment-1)

Name of Assignment	Continuous Assignment-1 (CA-1)
Course Name	Research Methodology (BTECH CTIS | Sem VII)
Faculty Name	Prof. Parmeshwari Aland (Asst. Professor / HOD-BCA)
Date / Deadline	September 2026 / October 2026
Problem Statement (SIH 26160)	AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework (NTRO)
Student Name & Group	**Aarohi Londhe** (Group: Sarvesh Bhamare, Aarohi Londhe, Pratik Shivekar)
Individual Focus Area	AI/ML & Deep Learning for Encrypted VPN Traffic Classification, Flow Fingerprinting, and Side-Channel Detection

1. Task 1 (LO1): Definition and Conceptualisation of Research and Its Purpose

Research is defined as a structured, methodical, and evidence-driven investigation undertaken to discover new facts, establish causal or statistical relationships, and develop innovative paradigms for solving complex technical challenges. Whereas everyday problem-solving relies on ad-hoc heuristics or existing operational manuals, academic and engineering research is governed by replicability, falsifiability, quantitative benchmarking, and critical synthesis of prior scholarly literature.

Within the domain of Network Security and Artificial Intelligence, research combines theoretical protocol modeling with empirical data science. When network traffic is encapsulated inside encrypted tunnels such as IPsec ESP, traditional deterministic inspection fails because payload bytes are computationally indistinguishable from random noise. Consequently, scientific inquiry in Encrypted Traffic Analysis (ETA) investigates how observable side-channel metadata—such as packet inter-arrival times (IAT), packet length sequences, burst dynamics, and directional flow entropies—can be modeled mathematically using deep neural networks and self-attention architectures. The purpose of this research is threefold: (1) Exploratory Research to extract invariant statistical and temporal signatures of IPsec IKEv1/IKEv2 and ESP flows across heterogeneous networks; (2) Empirical Research to evaluate deep learning and few-shot meta-learning classifiers on encrypted VPN datasets; and (3) Applied Systems Research to embed these models into an automated IPsec Security Assessment Framework.

2. Tasks 2 & 3 (LO2 & LO3): Research Area Selection and Topic Justification (300+ Words)

Broad Research Area: Encrypted Network Traffic Analytics, Machine Learning in Cybersecurity, and Protocol Security.
Specific Research Topic: AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework (Smart India Hackathon 2026 Problem Statement ID: 26160, National Technical Research Organisation - NTRO).

Justification of Relevance and Feasibility: With over 90% of global internet traffic now encrypted and remote work driving a 200% surge in Virtual Private Network adoption (Maghsoudlou et al., 2023; Gunasekara et al., 2026), security operations centers and national defense agencies face a critical visibility paradox. IPsec VPNs protect enterprise, military, and 5G/IoT infrastructures by encrypting payloads at the IP layer. However, this same encryption conceals protocol misconfigurations, unauthorized shadow-VPN tunnels, data exfiltration, and timing side-channel attacks (Moghimi et al., 2019). Furthermore, traditional Deep Packet Inspection (DPI) and port-based classifiers (checking UDP ports 500 and 4500 or IP protocol 50/51) are easily bypassed when VPN traffic is encapsulated over TCP port 443 or obfuscated by NAT-Traversal.

Narrowing the research topic to an AI-Powered IPsec VPN Protocol Analyzer is strongly justified because machine learning models—specifically 1D Convolutional Neural Networks (1D-CNNs), Transformers, and Multivariate Time-Series Attention networks—can accurately classify encrypted VPN sessions, infer underlying operational modes (Tunnel vs. Transport), and detect handshake timing anomalies without decrypting confidential user payloads. From a feasibility standpoint, benchmark datasets such as ISCXVPN2016, combined with programmable strongSwan/Open5GS testbeds, provide rich labeled PCAP traces to train and validate high-accuracy, lightweight AI models suitable for real-time deployment.

3. Task 4 (LO4): Design of Research Title with Detailed Explanation (200+ Words)

Designed Research Title: "AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework: Deep Learning and Metadata-Driven Encrypted Traffic Fingerprinting"

Title Explanation and Scope Alignment: Each phrase in the research title is deliberately chosen to reflect the methodology, target architecture, and analytical contributions of Aarohi Londhe's research module:
1. 'AI-Powered': Denotes the use of modern deep learning architectures—including Hardware-Aware 1D-CNNs (Chehade et al., 2025) and Self-Attention Meta-Learning (Gunasekara et al., 2026)—to automate protocol characterization.
2. 'IPsec VPN Protocol Analyzer': Specifies the target protocol stack comprising IKEv1/IKEv2 control-plane handshakes and ESP/AH data-plane flows across enterprise, IoT, and 5G networks.
3. 'Security Assessment Framework': Highlights the end-to-end diagnostic pipeline that translates raw packet captures into quantitative vulnerability scores and compliance reports.
4. 'Deep Learning and Metadata-Driven Encrypted Traffic Fingerprinting': Captures the core analytical mechanism—leveraging payload-agnostic traffic metadata (packet lengths, directions, burstiness, and inter-arrival times) to fingerprint encrypted IPsec tunnels, detect timing side-channels, and assess traffic analysis resistance.

4.1 Task 8 (LO8): Abstract Draft (Kept Same Across Group)

Virtual Private Networks (VPNs) serve as the foundational backbone for confidential, integrity-protected, and authenticated communication across untrusted public networks. Among existing tunneling architectures, Internet Protocol Security (IPsec) is extensively deployed across enterprise, government, defense, and cloud infrastructures. However, the real-world security posture of an IPsec deployment is highly sensitive to its negotiated cryptographic algorithms, Internet Key Exchange (IKEv1/IKEv2) state transitions, authentication mechanisms, Encapsulating Security Payload (ESP) parameters, operational modes (Tunnel versus Transport), and resilience against emerging post-quantum and side-channel threats. Misconfigurations, deprecated cipher suites, weak Diffie-Hellman groups, improper key management, or protocol implementation flaws frequently expose critical networks to downgrade, man-in-the-middle (MITM), and traffic fingerprinting attacks. Traditional packet analyzers such as Wireshark provide low-level packet visibility but require manual expert interpretation. This research proposes an AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework that combines automated IKE/ESP state-machine parsing, multivariate encrypted traffic metadata analysis, and Large Language Model (LLM)-assisted vulnerability reasoning to automatically fingerprint IPsec deployments, quantify cryptographic and operational risks, and generate actionable, standards-compliant remediation recommendations.

4.2 Task 8 (LO8): Research Introduction (400–500 Words)

The rapid proliferation of encrypted communication protocols—most notably Internet Protocol Security (IPsec), Transport Layer Security (TLS 1.3), and QUIC—has transformed the landscape of network security management. Across enterprise networks, government agencies, 5G mobile cores (non-3GPP N3IWF and O-RAN E2 interfaces), and Industrial Internet of Things (IIoT) deployments, IPsec serves as the primary mechanism for establishing Virtual Private Networks (VPNs) with strong confidentiality, data integrity, and peer authentication (Aragon et al., 2018; Atutxa et al., 2026; Perera et al., 2026). However, the operational security of an IPsec deployment depends heavily on harmonizing cryptographic negotiations, authentication infrastructures, operational modes (Tunnel versus Transport), and resistance to side-channel leakage.

A critical challenge in modern network defense is that traditional protocol analyzers rely either on static port mappings or payload-level Deep Packet Inspection (DPI). When inspecting IPsec traffic, DPI is rendered ineffective once the Internet Key Exchange (IKE) phase transitions to the Encapsulating Security Payload (ESP) data phase, as ESP encrypts both the transport header and payload. Moreover, byte-level deep learning models (such as ET-BERT) frequently overfit to static header artifacts or pseudorandom sequence numbers, suffering severe accuracy degradation when deployed across heterogeneous network domains or NAT-traversed VPN tunnels (Gunasekara et al., 2026). At the same time, subtle implementation flaws—such as non-constant-time ECDSA signature generation in hardware/firmware TPMs—can leak private authentication keys over remote strongSwan IPsec handshakes (Moghimi et al., 2019).

Consequently, cybersecurity analysts require an intelligent, automated framework capable of analyzing both the plaintext IKE negotiation parameters and the encrypted ESP flow dynamics. This study introduces an AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework. By modeling IPsec sessions as multivariate time series of packet sizes, inter-arrival times, and directional bursts, and coupling this classifier with a rule-based and LLM-enhanced cryptographic auditor, the proposed system automatically fingerprints IPsec implementations, evaluates susceptibility to traffic fingerprinting and timing attacks, identifies weak cipher suites, and generates actionable remediation strategies.

5. Task 5 (LO5): Preliminary Findings and Literature Review Summary (500+ Words)

To establish a rigorous foundation for the proposed framework, 25 peer-reviewed research papers were systematically collected and analyzed. Within the three-member research team, Aarohi Londhe conducted an in-depth investigation of Papers 10 through 17—which focus on constrained/5G IPsec profiles, authentication state management, remote timing side-channels, and deep learning for encrypted traffic analysis—while integrating cross-cutting findings from the entire 25-paper corpus.

1. Deep Learning and Metadata-Driven Encrypted Traffic Analysis (ETA): Gunasekara et al. (2026) [Paper 15] proposed GETA (Generalized Encrypted Traffic Analysis), demonstrating that byte-based transformers (such as ET-BERT) fail under VPN encapsulation and domain shifts because they overfit to plaintext header bytes. By modeling encrypted flows as multivariate time series of packet size, inter-arrival time (IAT), and direction using self-attention and meta-learning, GETA achieves superior accuracy across nine datasets including VPN classification. Complementing this, Chehade et al. (2025) [Paper 17] utilized Hardware-Aware Neural Architecture Search (HW-NAS) to design an INT8-quantized 1D-CNN that achieves 96.59% accuracy on the ISCX VPN-NonVPN dataset with only 88.26K parameters, proving that real-time encrypted VPN classification is feasible even on resource-constrained edge processors. Earlier foundational work by Dvir et al. (2016) [Paper 19] further established that burst-level statistical features can identify OS, browser, and application profiles inside encrypted VPN tunnels with over 85% accuracy.

2. Traffic Analysis Defenses and Adversarial Adaptation: Pulls and Witwer (2023) [Paper 16] introduced Maybenot, a Rust-based probabilistic state-machine framework that schedules adaptive padding and packet blocking to defend encrypted tunnels (such as Mullvad VPN's DAITA) against deep learning traffic analysis. Similarly, Sadeghzadeh et al. (2020) [Paper 5] developed Adversarial Website Adaptation (AWA), using deep learning transformers with secret random elements to reduce adversary fingerprinting accuracy over VPNs/IPsec to 19.52%.

3. Remote Timing Side-Channels and Authentication Vulnerabilities in IPsec: Moghimi et al. (2019) [Paper 13] uncovered TPM-FAIL (USENIX Security 2020), demonstrating that secret-dependent execution times during elliptic-curve signature generation allow a remote adversary to recover a strongSwan IPsec VPN server's 256-bit private key over the network in 45,000 authentication handshakes using lattice attacks. To harden VPN PKI authentication against quantum threats, Herzinger et al. (2025) [Paper 12] proposed a time-based stateful Hash-Based Signature (XMSS) Certificate Authority in OpenBSD iked that issues short-validity (4-hour) leaf certificates.

4. IPsec Deployment in IoT, 5G N3IWF, and O-RAN Architectures: Aragon et al. (2018) [Paper 10] designed the ACE IPsec profile for constrained IoT devices on Contiki OS, supporting both IKEv2 and direct SA provisioning. Atutxa et al. (2026) [Paper 11] integrated QKD keys into 5G Non-3GPP Interworking Function (N3IWF) IPsec tunnels, cutting SA establishment latency by up to 5.17%, while Perera et al. (2026) [Paper 14] experimentally proved that NIST ML-KEM-based IKEv2/IPsec on the 5G O-RAN E2 interface adds only 3–5 ms tunnel setup overhead.

Identified Research Gap: While existing ETA models classify VPN vs. non-VPN traffic with high accuracy, they act as black-box flow classifiers and do not correlate flow timing/metadata anomalies with IKEv1/IKEv2 cryptographic negotiation flaws, side-channel leakage risks, or automated security remediation.

6. Task 6 (LO6): Data Gathering, Organisation, and Proposed Methodology

In fulfillment of LO6 and Tasks 6–8, 25 peer-reviewed research papers were downloaded as full-text PDFs and cataloged in Reference_Papers_25_CA1.xlsx. Within the group of 3 (Sarvesh Bhamare, Aarohi Londhe, Pratik Shivekar), Aarohi Londhe analyzed Papers 10–17. The proposed AI-Powered IPsec Analyzer comprises four stages: (1) Multi-Source PCAP & Config Ingestion; (2) Deterministic IKEv1/IKEv2 State & Cryptographic Proposal Parser; (3) Multivariate Time-Series Deep Learning Encrypted ESP Flow & Side-Channel Classifier; and (4) LLM-Driven Quantitative Security Risk Scoring and Automated Remediation Synthesis.

7. Task 7 (LO7): Comprehensive Research Paper Outline (500+ Words)

Section 1.0: Abstract & Executive Summary — Scope, AI methodology, and key security assessment metrics.
Section 2.0: Introduction & Problem Statement — IPsec architecture (IKEv1/v2, ESP, AH), Tunnel vs. Transport mode risks, DPI blindness, NTRO Problem Statement 26160, and contributions of Aarohi Londhe.
Section 3.0: Literature Review & Threat Taxonomy — Critical synthesis of 25 papers covering VPN ecosystem scans, PQC/QKD migration (RFC 9242/9370, ML-KEM, XMSS), side-channel/DoS attacks (TPM-FAIL, Stealth MITM), and AI/LLM protocol analysis (GETA, HW-NAS 1D-CNN, SeriCrypt).
Section 4.0: System Architecture — IKEv1/v2 Handshake Dissector, Multivariate Encrypted Flow Classifier, and CDSL State Validator.

Section 5.0: Quantitative Security Risk Scoring & LLM Remediation — Multi-factor scoring across Confidentiality, Integrity, Authentication, Key Exchange, DoS/Side-Channel resilience, and Post-Quantum readiness.
Section 6.0: Experimental Evaluation — Multi-vendor IPsec testbed (strongSwan, Libreswan, OpenBSD iked, P4 BMv2) and ISCXVPN2016 benchmarks.
Section 7.0: Discussion & Viva Voce Insights — Operational scalability, adversarial traffic morphing defenses, and privacy preservation.
Section 8.0: Conclusion, Future Scope & APA References — Summary of contributions and 25+ APA citations.

8. Aarohi Londhe's Assigned Research Papers (Papers 10–17): Reference, Verbatim Abstract, Introduction & Literature Review

Paper 10 [Aarohi Londhe]: ACE of Spades in the IoT Security Game: A Flexible IPsec Security Profile for Access Control

1. APA Reference: Aragon, S., Tiloca, M., Maass, M., Hollick, M., & Raza, S. (2018). ACE of Spades in the IoT Security Game: A Flexible IPsec Security Profile for Access Control. 2018 IEEE Conference on Communications and Network Security (CNS), Beijing, China, 2018, pp. 1-9. https://doi.org/10.1109/CNS.2018.8433209 (arXiv:1808.04581) (Local File: Paper_10_1808.04581_ACE_of_Spades_in_the_IoT_Security_Game_A_Flexible.pdf)

2. Abstract (Verbatim / Kept Same): The Authentication and Authorization for Constrained Environments (ACE) framework provides fine-grained access control in the Internet of Things, where devices are resource-constrained and with limited connectivity. The ACE framework defines separate profiles to specify how exactly entities interact and what security and communication protocols to use. This paper presents the novel ACE IPsec profile, which specifies how a client establishes a secure IPsec channel with a resource server, contextually using the ACE framework to enforce authorized access to remote resources. The profile makes it possible to establish IPsec Security Associations, either through their direct provisioning or through the standard IKEv2 protocol. We provide the first Open Source implementation of the ACE IPsec profile for the Contiki OS and test it on the resource-constrained Zolertia Firefly platform. Our experimental performance evaluation confirms that the IPsec profile and its operating modes are affordable and deployable also on constrained IoT platforms.

3. Introduction: The Internet of Things (IoT) refers to network scenarios where billions of devices communicate over IP networks and are available on the Internet. This includes everyday objects and appliances, and has been constantly fostering a number of use cases and business opportunities, from sensor and actuator networks to smart buildings, from monitoring of critical infrastructures to controlled resource sharing. As more and more applications are being developed, the IoT is expected to have a huge impact on the way we live and work. At the same time, security plays a fundamental role, even during this transition process. In fact, ensuring security in IoT scenarios is of vital importance to counteract information breaches and service dysfunctions, which may result in severe performance degradation and privacy violations, or even threaten safety of people and infrastructures. Securing the IoT is thus vital to ensure its successful deployment and adoption. However, unlike in traditional networks, IoT devices are typically resource-constrained, i.e. equipped with limited resources. That is, they are scarce as to processing power, Fig. 1: IoT applications can secure their communications by using the ACE framework and the different security profiles. storage and energy availability, often being battery-powered. Besides, most IoT devices are wirelessly connected over lowpower and lossy networks, thus exhibiting limited connectivity and availability. Also, they often lack traditional user interfaces, and are likely deployed in unattended environments. As a result, protecting billions of IoT devices with traditional approaches is challenging, which fosters the development of novel security solutions suitable for the IoT. Yet, many of these solutions do not base on established standards a

4. Literature Review & Findings: [Venue/Publisher: IEEE Conference on Communications and Network Security (CNS 2018), IEEE | Methodology: Design and Contiki OS / Zolertia Firefly implementation of an OAuth 2.0-based ACE IPsec profile supporting IKEv2 and direct SA provisioning. | Key Findings: Proves that standardized IPsec Security Associations (SAs) and IKEv2 signaling can be adapted to resource-constrained IoT devices with low memory footprint. | Research Gap: Manual provisioning of ACE security profiles does not detect runtime protocol downgrade or replay anomalies.] II. RELATED WORK Different profiles have been proposed for the Authentication and Authorization for Constrained Environments (ACE) framework. In [5], Gerdes et al. describe the Datagram Transport Layer Security (DTLS) profile, which delegates the authorization and authentication of a Client device to the establishment of a DTLS session [6] between the Client and a Resource Server (RS). Specifically, DTLS can be used in the symmetric Pre-Shared Key (PSK) mode or the asymmetric Raw Public Key (RPK) mode. If the PSK mode is used, the successful establishment of a DTLS session also acts as a proof-of-possession (PoP) for the Client's PSK. In case the RPK mode is used, the Client is authenticated through its asymmetric public key. Finally, this profile uses the Constrained Application Protocol (CoAP) [17] over DTLS between Client and RS. The feasibility of securing CoAP messages with DTLS has been investigated in [1]. The OSCORE profile of ACE proposed by Seitz et al. [7] provides communication security between Client and RS by means of the Object Security for Constrained RESTful Environments (OSCORE) protocol [8]. OSCORE ensures request/response binding and selectively protects CoAP messages at the application layer, by using the compact CBOR Object Signing and Encryption (COSE) [18] based on the Concise Binary Object Representation (CBOR) [19] as data encoding format. This provides true end-to-end secure communication between Client and RS, even in the presence of (untrusted) intermediary CoAP proxies, which remain able to perform their intended operations (e.g. message cachin

Paper 11 [Aarohi Londhe]: IPsec based on Quantum Key Distribution: Adapting non-3GPP access to 5G Networks to the Quantum Era

1. APA Reference: Atutxa, A., Sanz, A., Salegi, E., González, G., Astorga, J., & Jacob, E. (2026). IPsec based on Quantum Key Distribution: Adapting non-3GPP access to 5G Networks to the Quantum Era. arXiv preprint arXiv:2603.24426. https://arxiv.org/abs/2603.24426 (Local File: Paper_11_2603.24426_IPsec_based_on_Quantum_Key_Distribution_Adapting_n.pdf)

2. Abstract (Verbatim / Kept Same): The advent of quantum computing will pose great challenges to the current communication systems, requiring essential changes in the establishment of security associations in traditional architectures. In this context, the multi-technological and heterogeneous nature of 5G networks makes it a challenging scenario for the introduction of quantum communications. Specifically, 5G networks support the unification of non-3GPP access technologies (i.e. Wi-Fi), which are secured through the IPsec protocol suite and the Non-3GPP Interworking Function (N3IWF) entity. These mechanisms leverage traditional public key cryptography and Diffie-Hellman key exchange mechanisms, which should be updated to quantum-safe standards. Therefore, in this paper we present the design and development of a Quantum Key Distribution (QKD) based non-3GPP access mechanism for 5G networks, integrating QKD keys with IPsec tunnel establishment. Besides, we also demonstrate the feasibility of the system by experimental validation in a testbed with commercial QKD equipment and an open-source 5G core implementation. Results show that the time required to complete the authentication and IPsec security association establishment is 4.62% faster than traditional cryptography PSK-based systems and 5.17% faster than the certificate-based system, while ensuring Information-Theoretic Security (ITS) of the QKD systems.

3. Introduction: Fifth-generation mobile networks (5G) have become a critical part of current communication infrastructures. These networks support a variety of applications, including ultrareliable and low-latency communications, as well as massive machine-type communications. From a security perspective, 5G relies on a combination of native 3GPP mechanisms and widely used transport and network security protocols that ensure confidentiality, authentication, and data integrity, such as TLS or IPsec [1]. These protocols currently build on public key primitives for authentication and key exchange, leveraging mechanisms such as Diffie-Hellman (D-H). With the advent of quantum computing, traditional publickey cryptographic algorithms may render vulnerable due to the development of the Shor's algorithm [2]. This potential paradigm shift in cryptography also presents risks to 5G networks. Consequently, advancing towards quantum-safe 5G deployments involves replacing or enforcing classical key establishment mechanisms with quantum-safe alternatives, such as Quantum Key Distribution (QKD). In this context, the segments between access and core sites are specially relevant, as they often span geographically separated locations where maintaining security guarantees is essential. Current 5G deployments typically protect these interfaces using IPsec tunnels established via IKEv2. The security of these tunnels relies on the symmetric keys negotiated by IKEv2, which are derived from classical Diffie-Hellman key exchange mechanisms and are therefore vulnerable to quantum attacks. Additionally, one of the key features introduced by 5G networks is the heterogeneity, bringing and integrating multitechnology systems within the same network. Thus, with the aim of incorporating additional networks beyond 3GP

4. Literature Review & Findings: [Venue/Publisher: arXiv Cryptography and Security (cs.CR) / 5G Security Testbed Study, 2026 | Methodology: Hardware testbed integration of commercial QKD keys into 5G Non-3GPP Interworking Function (N3IWF) IKEv2/IPsec tunnels over the NWu interface. | Key Findings: Reduces IKEv2 authentication and IPsec SA establishment time by 4.62% compared to PSK and 5.17% compared to PKI certificates while providing Information-Theoretic Security. | Research Gap: Focuses specifically on 5G N3IWF access tunnels; does not provide general-purpose AI traffic diagnostics for heterogeneous enterprise IPsec gateways.] II. RELATEDWORK The integration of quantum communications into the security of optical fronthaul and backbone networks has been a recent topic in the research community. Current 5G networks, as well as next generation networks (i.e. 6G), heavily rely on the heterogeneity and multi-technology, inevitably increasing the complexity of transitioning to security in the quantum era. Early work on 5G and beyond 5G networks focused on embedding QKD equipment in the fronthaul. In [4], Zavitsanos et al. propose to secure eCPRI transport layers using QKD for key exchanges between the centralized Baseband Unit (BBU) and distributed 5G terminal nodes. Similarly, in [5] Zhang et al. present an architecture that enables the coexistence of quantum and classical channel leveraging QKD with Wavelength Division Multiplexing (WDM) for 6G fronthaul networks. These coexistence solutions based on WDM systems can also encrypt the data channels at line rate using keys generated by the QKD equipment, as demonstrated by Sanz et al. [6], achieving successful long-range encryption for fronthaul deployments. Additional works focus on embedding QKD into the security protocols used in each network segment of 5G networks, such as TLS and IPsec. For instance, Atutxa et al. [7] integrate and experimentally validate a QKD infrastructure in the 5G core, where TLS-protected service-based interfaces between network functions are supplied with QKD keys. In a complementary direction, Wright et al. [8] extend the 5G network slicing model by introducing a Software-Defined Networking (SDN)-based orchestrator capable

Paper 12 [Aarohi Londhe]: Time-Based State-Management of Hash-Based Signature CAs for VPN-Authentication

1. APA Reference: Herzinger, D., Heise, L., Loebenberger, D., & Söllner, M. (2025). Time-Based State-Management of Hash-Based Signature CAs for VPN-Authentication. arXiv preprint arXiv:2509.11695. https://arxiv.org/abs/2509.11695 (Local File: Paper_12_2509.11695_Time_Based_State_Management_of_Hash_Based_Signatur.pdf)

2. Abstract (Verbatim / Kept Same): Advances in quantum computing necessitate migrating the entire technology stack to post-quantum cryptography. This includes IPsec-based VPN connection authentication. Although there is an RFC draft for post-quantum authentication in this setting, the draft does not consider (stateful) hash-based signatures despite their small signature size and trusted long-term security. We propose a design with time-based state-management that assigns VPN devices a certificate authority (CA) based on the hash-based signature scheme XMSS. The CA then issues leaf certificates which are based on classical cryptography but have a short validity time, e. g., four hours. It is to be expected that even large quantum computers will take significantly longer to break the cryptography, making the design quantum-secure. We propose strategies to make the timekeeping more resilient to faults and tampering, as well as strategies to recognize a wrong system time, minimize its potential damage, and quickly recover. The result is an OpenBSD implementation of a quantum-safe and, regarding the leaf certificates, highly flexible VPN authentication design that requires significantly less bandwidth and computational resources compared to existing alternatives.

3. Introduction: Our world today is highly reliant on secure encryption. However, the advances in the field of quantum computing threaten to break all currently prevalent public key cryptography [3], an essential building block for cryptographic systems establishing secure connections, e.g., VPNs. A Quantum Computer (QC) with a sufficient amount of logical qubits can break the underlying problems of factorization or discrete log based public key cryptography in (quantum) polynomial time[19].WecallaQCwithsufficientpoweraCryptographicallyRelevantQuantum Computer (CRQC). Estimates regarding when a CRQC will exist depend 2 D. Herzinger et al. Fig.1.Historical estimates with comparable physical assumptions of the physical qubit cost of factoring 2048 bit RSA integers [8]. on both, the number of logical qubits that are necessary to break the scheme, and also when this number of qubits will be achieved in real world QCs. Figure 1 depicts the trend of those forecasts regarding the amount of necessary physical qubits to factorize a 2048 bit RSA integer [8]. A report by the global risk institute indicates a similarly worrying trend with a significant probability of QCs being able to break 2048 bit RSA in 24 hours during the next decade [14]. In response to this development, the US National Institute of Standards and Technology (NIST) designates most currently used public key algorithms as deprecated or even disallowed after2030–2035[13]. Instead, organizations should transition to algorithms of the field of post-quantum cryptography (PQC) which are expected to be secure also against strong quantum computers. European agencies,however,promoteasimilarscheduleforintroducingPQCwithastronger focus on hybrid solutions that do not require as much trust in the long-term security of the new PQC algorithms

4. Literature Review & Findings: [Venue/Publisher: arXiv Cryptography and Security (cs.CR), 2025 | Methodology: Time-based stateful Hash-Based Signature (XMSS) Certificate Authority architecture implemented in OpenBSD iked for quantum-safe IKEv2 VPN authentication. | Key Findings: Combines long-term quantum-secure XMSS root CAs with short-lived (4-hour) leaf certificates, drastically reducing IKEv2 authentication bandwidth vs. SLH-DSA. | Research Gap: Vulnerable to subtle system clock manipulation if timekeeping faults occur; requires automated certificate and state-consistency monitoring.] 2 Related Work Severalauthorshaveaddresseddifferentapproachesforsecurestate-management of S-HBS, including time-based approaches, and also the integration of PQC into IPsec. We now explore the respective state of the art and lay the groundwork for understanding our design. Wiggers et al. discuss in their current Internet Draft various aspects and security considerations of state and backup management for S-HBS [22]. They took a general approach and discussed various different techniques, including timebased management, which we focus on. They propose a system in which available signatures are reserved for certain time windows, more specifically eight windows containing 128 signatures for a total of 1024 available signatures. They identified several challenges regarding its time-based state-management approach: 4 D. Herzinger et al. –Accurate time keeping on the signing devices is essential. This is a serious challenge rooted in computer engineering. –The index/state of the S-HBS can only increase, stepping back could cause a second signing with the same OTS key, which must not happen. –The number of created signatures in each window has to be tracked to guarantee that no more key pair than allowed is used. –The signing device has to maintain a proper state-management in general, i.e., update the state accordingly etc. –The time keeping on the signing device has to be secure against malicious manipulation. –In case of a replacement of the signing device the new device has to be in sync with the initial time keeping. –An additional rate limit of the signatures as well as a se

Paper 13 [Aarohi Londhe]: TPM-FAIL: TPM meets Timing and Lattice Attacks

1. APA Reference: Moghimi, D., Sunar, B., Eisenbarth, T., & Heninger, N. (2019). TPM-FAIL: TPM meets Timing and Lattice Attacks. arXiv preprint arXiv:1911.05673. https://arxiv.org/abs/1911.05673 (Local File: Paper_13_1911.05673_TPM_FAIL_TPM_meets_Timing_and_Lattice_Attacks.pdf)

2. Abstract (Verbatim / Kept Same): Trusted Platform Module (TPM) serves as a hardware-based root of trust that protects cryptographic keys from privileged system and physical adversaries. In this work, we perform a black-box timing analysis of TPM 2.0 devices deployed on commodity computers. Our analysis reveals that some of these devices feature secret-dependent execution times during signature generation based on elliptic curves. In particular, we discovered timing leakage on an Intel firmware-based TPM as well as a hardware TPM. We show how this information allows an attacker to apply lattice techniques to recover 256-bit private keys for ECDSA and ECSchnorr signatures. On Intel fTPM, our key recovery succeeds after about 1,300 observations and in less than two minutes. Similarly, we extract the private ECDSA key from a hardware TPM manufactured by STMicroelectronics, which is certified at Common Criteria (CC) EAL 4+, after fewer than 40,000 observations. We further highlight the impact of these vulnerabilities by demonstrating a remote attack against a StrongSwan IPsec VPN that uses a TPM to generate the digital signatures for authentication. In this attack, the remote client recovers the server's private authentication key by timing only 45,000 authentication handshakes via a network connection. The vulnerabilities we have uncovered emphasize the difficulty of correctly implementing known constant-time techniques, and show the importance of evolutionary testing and transparent evaluation of cryptographic implementations. Even certified devices that claim resistance against attacks require additional scrutiny by the community and industry, as we learn more about these attacks.

3. Introduction: Hardware support for trusted computing has been proposed based on trusted execution environments (TEE) and secure elements such as the Trusted Platform Module (TPM) [40]. Computer manufacturers have been deploying TPMs on desktop workstations, laptops, and servers for over a decade. With a TPM device attached to the computer, the root of trust can be executed in a separate hardened cryptographic core, which prevents even a fully compromised OS from revealing credentials or keys to adversaries. TPM 2.0, the latest standard, is deployed in almost all modern computers and is required by some core security services [38]. TPM 2.0 supports multiple signature schemes based on elliptic curves [63]. TPMs were originally designed as separate hardware modules, but new demands have resulted in software-based implementations. The physical separation of the TPM from the CPU is an asset for protection against system-leval adversaries [3]. However, its lightweight design and low-bandwidth bus connection prevents the TPM from being used as a secure cryptographic co-processor for high-throughput applications. TEE technologies such as ARM TrustZone [2] are a more recent approach to bringing trusted execution right into the CPU, at minimal performance loss. Firmware TPMs (fTPM) can run entirely in software within a TEE like ARM Trustzone [48]. In a cloud environment, a software-virtualized TPM device will be executed within the trust boundary of the hypervisor [23, 39, 46]. In this case, user applications still benefit from the defense against attacks on the guest OS. Virtual TPMs may or may not rely on a physically present TPM hardware. Intel Platform Trust Technology (PTT), introduced in Haswell processors, is based on fTPM and follows a hybrid hardware/software approach to implement the

4. Literature Review & Findings: [Venue/Publisher: 29th USENIX Security Symposium (USENIX Security 2020), USENIX Association | Methodology: Black-box timing side-channel analysis and lattice-based Hidden Number Problem (HNP) key recovery against TPM 2.0 ECDSA/ECSchnorr used by strongSwan IPsec VPNs. | Key Findings: Recovers 256-bit private authentication keys from a remote strongSwan IPsec VPN server over the network in ~45,000 IKEv2 authentication handshakes. | Research Gap: Demonstrates that even hardware-backed IPsec VPNs leak secret keys via handshake timing; requires AI-based handshake timing anomaly detection.] 2 Background 2.1 Trusted Platform Module TPMs are secure elements which are typically dedicated physical chips with Common Criteria certification at EAL 4 and higher, and thus provide a very high level of security assurance for the services they offer [12]. As shown in Figure 1, the TPM, including components like cryptographic engines, forms the root of trust. On a commodity computer, the host processor is connected to the TPM via a standard communication interface [59]. For trusted execution of cryptographic protocols, applications can request that the OS interact with the TPM device and use various cryptographic engines that support hash functions, encryption, and digital signatures. The TPM also contains non-volatile memory for secure storage of cryptographic parameters and configurations. As discussed in Section 5.3, for instance, a Virtual Private Network (VPN) 1https://github.com/tpm2-software/tpm2-tools commit c66e4f0 2https://github.com/tpm2-software/tpm2-tss commit 443455b TrustedUntrusted TPM PCR Registers Crypto Engine Random Number Generator Execution Engine Volatile Memory Non-volatileMemory Host CPU Main Memory System Software Applications Remote Attestation Request Figure 1: The trusted components of a TPM include the PCR registers, crypto engine, and random number generator. Other hardware components, system software, and applications are considered untrusted. application can use the TPM to securely store authentication keys and to perform authentication without direct access to the private key. TPM also supports remote attestation, in which the TPM will gen

Paper 14 [Aarohi Londhe]: Towards Quantum-Safe O-RAN -- Experimental Evaluation of ML-KEM-Based IPsec on the E2 Interface

1. APA Reference: Perera, M., Mackay, M., Eiza, M. H., Raschellà, A., Shone, N., & Maheshwari, M. K. (2026). Towards Quantum-Safe O-RAN -- Experimental Evaluation of ML-KEM-Based IPsec on the E2 Interface. arXiv preprint arXiv:2601.20378. https://arxiv.org/abs/2601.20378 (Local File: Paper_14_2601.20378_Towards_Quantum_Safe_O_RAN_Experimental_Evaluation.pdf)

2. Abstract (Verbatim / Kept Same): As Open Radio Access Network (O-RAN) deployments expand and adversaries adopt 'store-now, decrypt-later' strategies, operators need empirical data on the cost of migrating critical control interfaces to post-quantum cryptography (PQC). This paper experimentally evaluates the impact of integrating a NIST-aligned module-lattice KEM (ML-KEM, CRYSTALS-Kyber) into IKEv2/IPsec protecting the E2 interface between the 5G Node B (gNB) and the Near-Real-Time RAN Intelligent Controller (Near-RT RIC). Using an open-source testbed built from srsRAN, Open5GS, FlexRIC and strongSwan (with liboqs), we compare three configurations: no IPsec, classical ECDH-based IPsec, and ML-KEM-based IPsec. The study focuses on IPsec tunnel-setup latency and the runtime behaviour of Near-RT RIC xApps under realistic signalling workloads. Results from repeated, automated runs show that ML-KEM integration adds a small overhead to tunnel establishment, which is approximately 3~5 ms in comparison to classical IPsec, while xApp operation and RIC control loops remain stable in our experiments. These findings indicate that ML-KEM based IPsec on the E2 interface is practically feasible and inform quantum-safe migration strategies for O-RAN deployments.

3. Introduction: Secure communication protocols are fundamental to trustworthy mobile networks in increasingly complex Fifth Generation (5G) deployments. Open Radio Access Networks (O-RANs), aka Open-RANs, have emerged as a key architectural paradigm for 5G and beyond, enabling service heterogeneity, multi -vendor interoperability, virtuali sation, and AI -driven intelligence. Specifically, O -RAN defines a fully virtualised RAN in which base station functions are disaggregated into an open Central Unit (CU) and Distributed Unit (DU) and deployed within a cloud environment or data centre, along with a hardware -based Radio Unit (RU) interconnected through open interfaces. These components are managed and optimised by centralised control entities, namely the RAN Intelligence Controllers (RICs), which maintain a comprehensive view of the network [1]. This disaggregation, however, significantly enlarges the attack surface, making interface protection a central security concern [2]. Therefore, the O -RAN Alliance Working Group (WG11), has positioned “security by design” as a core principle, specifying mandatory controls such as TLS and IPsec for inter-node communication [3]. Among these, the E2 interface linking the 5G Node B (gNB)’s CU/DU with the Near -RT RIC is particularly critical as it carries near -real-time control and telemetry data and is subject to stringent timing constraints. WG11 mandates IPsec on E2 to ensure confidentiality and integrity, but stronger cryptographic mechanisms may impact latency and resource usage in ways that are not yet fully explored [3, 4]. In parallel, advances in quantum computing threaten widely deployed public -key primitives such as Rivest–Shamir– Adleman (RSA), Elliptic Curve Diffie–Hellman (EC)DH, and Elliptic Curve Cryptography (ECC) thus, the lon

4. Literature Review & Findings: [Venue/Publisher: arXiv Cryptography and Security (cs.CR) / O-RAN Security Evaluation, 2026 | Methodology: Empirical evaluation of NIST FIPS 203 ML-KEM (CRYSTALS-Kyber) in strongSwan IKEv2/IPsec over the 5G O-RAN E2 interface using srsRAN, Open5GS, and FlexRIC. | Key Findings: Shows ML-KEM-512/768/1024 adds only 3-5 ms overhead to IPsec tunnel setup while maintaining stable Near-RT RIC xApp control loops. | Research Gap: Evaluated in controlled testbeds; production networks need automated continuous assessment of IKEv2 KEM negotiation and fragmentation behavior.] II. LITERATURE REVIEW This literature review introduces O -RAN architecture including its interfaces and security, and the latest works on PQC evaluation in network protocols (IPsec/TLS) , and recent industry and standards activity towards quantum -safe telecoms. A. O-RAN Architecture The O-RAN Alliance rearchitects the 5G RAN by extending the 3GPP NG -RAN “split” gNB into a fully disaggregated, cloud-native, and programmable platform. Building on the logical separation of the gNB into CU, DU, and RU in 3GPP Release 15, O -RAN renames these elements O -CU, O -DU, and O-RU, respectively, and standardises open, interoperable interfaces between them. This approach inherits ideas from Cloud (C -RAN) and Virtualised RAN (V -RAN) and goes further by mandating openness of interfaces and embedding intelligence as a first -class design goal [14]. It is now the reference architecture for most academic and industrial work on Open RAN [15]. The O-RAN architecture proposed by the O-RAN Alliance is depicted in Fig. 1. Figure 1: Logical Architecture of O-RAN [16] At the radio side, the O -CU serves as the intelligent control hub that acts as the bridge between the RAN and the core network. It handles higher -layer protocol processing up to Packet Data Convergence Protocol (PDCP), terminates the NG interface towards the 5G core, and coordinates with the ODU over the 3GPP F1-C/F1-U interfaces. The O-DU performs mid-layer RAN functions such as Radio Link Control (RLC), MAC, and upper PHY, and aggregates multiple O-RUs via the Open Fronthaul interface. The O -RU is responsible for Radio Frequ

Paper 15 [Aarohi Londhe]: GETA: Generalized Encrypted Traffic Analysis

1. APA Reference: Gunasekara, R., Masood, R., & Kanhere, S. (2026). GETA: Generalized Encrypted Traffic Analysis. arXiv preprint arXiv:2605.31277. https://arxiv.org/abs/2605.31277 (Local File: Paper_15_2605.31277_GETA_Generalized_Encrypted_Traffic_Analysis.pdf)

2. Abstract (Verbatim / Kept Same): Traditional traffic analysis is being fundamentally challenged by the rapid adoption of encryption, tunnelling, and privacy-preserving protocols, which increasingly obscure packet payloads and limit the usefulness of Deep Packet Inspection (DPI). Although machine learning has advanced encrypted traffic analysis, existing approaches often remain tied to protocol-specific header features, depend on large labelled datasets, and degrade when deployed across heterogeneous network environments. We present GETA, a protocol-agnostic framework for encrypted traffic analysis that models network flows as multivariate time series using only traffic metadata, thereby avoiding reliance on packet payloads or header semantics. GETA combines meta-learning, embedding refinement, and self-attention to support few-shot adaptation to previously unseen domains with minimal labelled data. Across nine public datasets spanning application identification, VPN traffic classification, IoT device fingerprinting, and attack detection, GETA consistently outperforms state-of-the-art baselines. These results show that GETA offers a practical and generalisable foundation for robust traffic analysis in modern encrypted networks.

3. Introduction: The widespread adoption of encryption protocols such as TLS, QUIC, and IPsec enhances the confidentiality and integrity of network communications but limits the effectiveness of traditional monitoring methods, such as deep packet inspection (DPI), that rely on payload inspection. Consequently, conventional traffic analysis struggles to characterize and classify encrypted flows. In contrast, machine learning (ML) approaches exploit observable flow features such as packet sizes, interarrival times, and directions to analyze traffic without accessing payloads [1], [2], enabling scalable and privacy-preserving visibility into encrypted networks. Encrypted traffic analysis (ETA) has emerged as a key paradigm for understanding network behavior and detecting malicious activity through inference from traffic patterns, flow dynamics, and statistical characteristics [3]. With over 90% of web traffic now encrypted [4], the ability to analyze traffic flows without inspecting packet contents has become essential for both research and operational network security. Advances in ML and deep learning (DL) have enhanced ETA by enabling automated extraction of complex, non-linear patterns in encrypted traffic, improving classification and behavioral modeling [5]. However, several approaches still rely on packet header information [6], [7], a major limitation in scenarios involving VPNs, tunnels, or privacy-preserving networks where headers are minimal, encrypted, or entirely obfuscated, leading to degraded accuracy and robustness across diverse environments [8]. A key challenge for ML-based ETA is its reliance on large labeled datasets. While supervised learning performs well in controlled settings, it demands extensive data collection and manual labeling, limiting scalability in real-worl

4. Literature Review & Findings: [Venue/Publisher: arXiv Cryptography and Security (cs.CR), 2026 | Methodology: GETA protocol-agnostic encrypted traffic analysis framework combining multivariate time-series metadata (size, IAT, direction), self-attention, and meta-learning. | Key Findings: Outperforms byte-based models (like ET-BERT) across 9 public datasets including VPN traffic classification and attack detection with few-shot adaptation. | Research Gap: Classifies encrypted flow behaviors effectively but does not parse unencrypted IKE handshake proposals to recommend cryptographic configuration fixes.] II. RELATEDWORK We organize the related work into three areas aligned with our methodology. First, we review deep learning methods for ETA, followed by Few-Shot/Zero-Shot Learning approaches that address data scarcity. Next, we examine Time Series Modeling, which motivates our use of structured traffic metadata. A. Encrypted Traffic Analysis (ETA) ETA seeks to infer application, device, or behavioral information from encrypted network flows without accessing payloads. Recent transformer- and LLM-inspired methods operate directly on raw packet bytes [7], [12], [13]. For instance, ET-BERT [7] applies a BERT-style model to Ethernet packet bytes, achieving high accuracy but requiring extensive pretraining and showing poor generalization in constrained settings such as VPNs or non-TCP traffic [8]. These limitations stem from a reliance on byte-level patterns and plaintext headers, which may be obfuscated or unavailable in practice. Furthermore, [14] shows that byte-based models frequently overfit to pseudorandom fields such as Sequence/Ack numbers, learning spurious correlations rather than true traffic semantics. Metadata-based ETA methods, which predate byte-based deep learning, instead rely on flow-level features such as packet size, direction, and timing [15], [16]. While these methods offer greater interpretability and robustness to encryption or protocol variation, they traditionally require large labeled datasets and remain domain-specific [8], [17], [18].As byte-based methods remain vulnerable to obfuscation and domain shifts, revisiting metadata-based analysis with mode

Paper 16 [Aarohi Londhe]: Maybenot: A Framework for Traffic Analysis Defenses

1. APA Reference: Pulls, T., & Witwer, E. (2023). Maybenot: A Framework for Traffic Analysis Defenses. arXiv preprint arXiv:2304.09510. https://arxiv.org/abs/2304.09510 (Local File:

Paper_16_2304.09510_Maybenot_A_Framework_for_Traffic_Analysis_Defenses.pdf)

2. Abstract (Verbatim / Kept Same): End-to-end encryption is a powerful tool for protecting the privacy of Internet users. Together with the increasing use of technologies such as Tor, VPNs, and encrypted messaging, it is becoming increasingly difficult for network adversaries to monitor and censor Internet traffic. One remaining avenue for adversaries is traffic analysis: the analysis of patterns in encrypted traffic to infer information about the users and their activities. Recent improvements using deep learning have made traffic analysis attacks more effective than ever before. We present Maybenot, a framework for traffic analysis defenses. Maybenot is designed to be easy to use and integrate into existing end-to-end encrypted protocols. It is implemented in the Rust programming language as a crate (library), together with a simulator to further the development of defenses. Defenses in Maybenot are expressed as probabilistic state machines that schedule actions to inject padding or block outgoing traffic. Maybenot is an evolution from the Tor Circuit Padding Framework by Perry and Kadianakis, designed to support a wide range of protocols and use cases.

3. Introduction: After decades of effort, end-to-end encryption is becoming ubiquitous, with modern protocols such as QUIC [33], HTTP/3 [7], TLS 1.3 [61], DoH [28], DoQ [32], MLS [4], and more encrypted by default. In parallel, technologies that provide unlinkability between end users and their IP addresses under various threat models are being increasingly adopted, such as Tor [16], VPNs [17, 18, 60], and Apple's iCloud Private Relay [3, 49]. Together, these two trends are making it harder for network operators to detect and block malicious traffic and for attackers to identify and target specific users. The final frontier is traffic analysis: inferences based on the metadata of encrypted traffic. While traffic analysis attacks have been studied for a long time [13, 39, 43, 67, 76], we see relatively few deployed real-world defenses. Most defenses can be found in technologies aiming for user–IP address unlinkability, such as Tor [16]; however, they remain relatively modest in induced overhead [52], limiting their efficacy against many attacks. Recent protocol standards do support essential building blocks and note that they may be used to defend against traffic analysis (cf. support for the PADDING frame in QUIC [33, §21.14] and reflections on traffic analysis in TLS 1.3 [61, §3]), yet they do not mandate any specific use. There are several reasons for the lack of deployed traffic analysis defenses. For one, the negative performance impact of padding and added latency is a significant concern: with high costs, the benefits must be very clear. This is compounded by the fact that the landscape of traffic analysis attacks and defenses is rapidly evolving, driven by advances in deep learning and artificial intelligence [42]. Additionally, one must recognize that it has taken decades to ach

4. Literature Review & Findings: [Venue/Publisher: Proceedings on Privacy Enhancing Technologies / WPES (ACM), 2023/2024 | Methodology: Rust-based probabilistic state-machine framework (Maybenot) for scheduling adaptive packet padding and blocking against deep learning traffic analysis in VPNs. | Key Findings: Successfully deployed in production (Mullvad VPN DAITA over WireGuard) to counter AI-guided encrypted traffic fingerprinting. | Research Gap: Introduces bandwidth and latency overhead that must be balanced against enterprise SLA requirements.] 7 Related Work Maybenot is based on the Tor Circuit Padding Framework [51, 52] by Perry and Kadianakis. The Tor Circuit Padding Framework is, in turn, a generalization of WTF-PAD [37] by Juárez et al., a website fingerprinting defense based on the concept of Adaptive Padding by Shmatikov and Wang [67]. 7.1 Tor Circuit Padding Framework As the name suggests, the Tor Circuit Padding Framework is a framework for implementing padding in Tor circuits. It is closely integrated into Tor's C codebase. Clients can negotiate fixed hardcoded "padding machines" with relays in the network. At the time of writing, while the framework is deployed as part of Tor, only two small padding machines are active in the network [52]. Their goal is to hide the setup of client-side onion service circuits, making them produce the same sequence of cells as non-onion circuits. Comparing Maybenot to the Tor Circuit Padding Framework, the main differences are that Maybenot is designed to be integrated into a wide range of protocols, is written in Rust as a library, and includes a richer set of features. The Tor Circuit Padding Framework is closely tied to Tor and supports negotiation of padding machines, which Maybenot considers out of scope. Maybenot machines support probabilistic transitions, padding, blocking, and associated bypass/replace flags, counters, timers, and signaling. Maybenot also has more distributions but removes support for histograms. Finally, Maybenot does not support RTT-based estimates as offsets for timers. RTT-based estimates and histograms are excluded due to a lack of identified

Paper 17 [Aarohi Londhe]: Energy-Efficient Deep Learning for Traffic Classification on Microcontrollers

1. APA Reference: Chehade, A., Ragusa, E., Gastaldo, P., & Zunino, R. (2025). Energy-Efficient Deep Learning for Traffic Classification on Microcontrollers. arXiv preprint arXiv:2506.10851. https://arxiv.org/abs/2506.10851 (Local File: Paper_17_2506.10851_Energy_Efficient_Deep_Learning_for_Traffic_Classif.pdf)

2. Abstract (Verbatim / Kept Same): In this paper, we present a practical deep learning (DL) approach for energy-efficient traffic classification (TC) on resource-limited microcontrollers, which are widely used in IoT-based smart

systems and communication networks. Our objective is to balance accuracy, computational efficiency, and real-world deployability. To that end, we develop a lightweight 1D-CNN, optimized via hardware-aware neural architecture search (HW-NAS), which achieves 96.59% accuracy on the ISCX VPN-NonVPN dataset with only 88.26K parameters, a 20.12K maximum tensor size, and 10.08M floating-point operations (FLOPs). Moreover, it generalizes across various TC tasks, with accuracies ranging from 94% to 99%. To enable deployment, the model is quantized to INT8, suffering only a marginal 1-2% accuracy drop relative to its Float32 counterpart. We evaluate real-world inference performance on two microcontrollers: the high-performance STM32F746G-DISCO and the cost-sensitive Nucleo-F401RE. The deployed model achieves inference latencies of 31.43ms and 115.40ms, with energy consumption of 7.86 mJ and 29.10 mJ per inference, respectively. These results demonstrate the feasibility of on-device encrypted traffic analysis, paving the way for scalable, low-power IoT security solutions.

3. Introduction: Traffic classification (TC) plays a critical role in ensuring the security and efficiency of modern networks [1]. It enables key tasks such as intrusion detection, network optimization, and resource allocation [2]. Yet, the rise of encryption has rendered traditional methods like Deep Packet Inspection (DPI) and port-based classification ineffective, as they rely on payload visibility or static port mappings [3], [4]. This challenge is further compounded by the reliance on cloud-based solutions, which require transferring raw traffic data to centralized servers for analysis [5]. Such cloud-based approaches introduce significant drawbacks, including increased latency, high bandwidth consumption, and privacy concerns. Meanwhile, microcontrollers—widely used in IoT and embedded systems—are becoming integral to connected infrastructures, driving demand for efficient on-device processing. This work addresses these challenges by targeting microcontroller-based IoT devices as platforms for performing local TC. These systems offer real-time decision-making while minimizing network overhead and privacy risks [5]. However, microcontrollers are resource-constrained, with limitations on This work was partially supported by project SERICS (PE00000014) under the MUR National Recovery and Resilience Plan funded by the European Union - NextGenerationEU. memory, computational power, and energy efficiency. These factors necessitate tailored optimizations to ensure effective classification within strict hardware limits. Accordingly, network TC can occur at packet, flow, or session granularity [6]. Packets, as the smallest transmission units, carry limited context when analyzed alone. Flows group packets unidirectionally for broader context but may still miss complete endpoint interactions

4. Literature Review & Findings: [Venue/Publisher: IEEE Symposium on Computers and Communications (IEEE ISCC 2025), IEEE | Methodology: Hardware-Aware Neural Architecture Search (HW-NAS) 1D-CNN quantized to INT8 for encrypted VPN vs. Non-VPN session classification on STM32 microcontrollers. | Key Findings: Achieves 96.59% accuracy on the ISCX VPN-NonVPN dataset with only 88.26K parameters and 31.43 ms inference latency on embedded hardware. | Research Gap: Classify traffic into VPN/non-VPN categories but does not inspect IPsec security associations or evaluate cipher suite weaknesses.] II. RELATED WORK A. Conventional Traffic Classification Techniques Early TC methods relied on port-based classification and Deep Packet Inspection (DPI). Port-based classification maps network traffic to applications using fixed port numbers. However, modern networks, with their use of dynamic and private ports, port obfuscation, and HTTP tunneling, have rendered this approach largely ineffective as noted in [4]. DPI, which examines packet payloads for patterns or signatures, was effective for unencrypted traffic but is unsuitable for encrypted communication, where payloads are inaccessible. Additionally, DPI is resource-intensive and raises privacy concerns, limiting its feasibility in contemporary environments [4], [13]. B. Machine Learning Techniques Machine learning (ML) has provided an alternative to conventional methods by leveraging statistical patterns in network traffic to classify encrypted communication without inspecting packet payloads. Early ML-based approaches used models such as Na ¨■ve Bayes, Random Forests, and k-NN to classify traffic based on handcrafted features like packet size and flow duration, as in [12], [14], [15]. While these methods showed reasonable accuracy, they required significant manual effort for feature selection and engineering to adapt to evolving network traffic patterns, limiting their scalability and robustness. DL has advanced TC by automating feature extraction, enabling the analysis of raw network traffic without manual feature engineering. Techniques such as convolutional neural networks (CNNs) and recurrent neural networks (RNN

9. Consolidated APA 7th Edition Reference List (All 25 Papers)

[01 - Sarvesh Bhamare] Marksteiner, S., Rainer, B., & Maurhart, O. (2018). On the Resilience of a QKD Key Synchronization Protocol for IPsec. International Journal on Advances in Security, vol 9 no 3 & 4, year 2016. https://arxiv.org/abs/1801.01710

[02 - Sarvesh Bhamare] Maghsoudlou, A., Vermeulen, L., Poese, I., & Gasser, O. (2023). Characterizing the VPN Ecosystem in the Wild. Proceedings of the Passive and Active Measurement Conference 2023 (PAM '23). https://arxiv.org/abs/2302.06566

[03 - Sarvesh Bhamare] Blanco-Romero, J., García, P. O., Sobral-Blanco, D., Mendoza, F. A., Vilas, A. F., & Fernández-Veiga, M. (2025). Hybrid Quantum Security for IPsec. arXiv preprint arXiv:2507.09288. https://arxiv.org/abs/2507.09288

[04 - Sarvesh Bhamare] Alam, S. (2026). Quantum-Ready Secure WAN: A Risk Assessment and Migration Framework. arXiv preprint arXiv:2609.26225. https://arxiv.org/abs/2609.26225

[05 - Sarvesh Bhamare] Sadeghzadeh, A. M., Tajali, B., & Jalili, R. (2020). AWA: Adversarial Website Adaptation. arXiv preprint arXiv:2012.10832. https://arxiv.org/abs/2012.10832

[06 - Sarvesh Bhamare] Mostafa, M., Kalam, A. A. E., & Fraboul, C. (2009). Q-ESP: a QoS-compliant Security Protocol to enrich IPSec Framework. Third IEEE / IFIP International Conference on New Technologies, Mobility and Security, Cairo : Egypt (2009). https://arxiv.org/abs/0911.4034

[07 - Sarvesh Bhamare] Hauser, F., Häberle, M., Schmidt, M., & Menth, M. (2019). P4-IPsec: Site-to-Site and Host-to-Site VPN with IPsec in P4-Based SDN. arXiv preprint arXiv:1907.03593. https://doi.org/10.1109/ACCESS.2020.3012738

[08 - Sarvesh Bhamare] Jumakhan, H., & Mirzaeinia, A. (2024). Wireguard: An Efficient Solution for Securing IoT Device Connectivity. arXiv preprint arXiv:2402.02093. https://arxiv.org/abs/2402.02093

[09 - Sarvesh Bhamare] Mallick, T., Kundu, A., & Kompella, R. (2026). Study of Post Quantum status of Widely Used Protocols. arXiv preprint arXiv:2603.28728. https://arxiv.org/abs/2603.28728

[10 - Aarohi Londhe] Aragon, S., Tiloca, M., Maass, M., Hollick, M., & Raza, S. (2018). ACE of Spades in the IoT Security Game: A Flexible IPsec Security Profile for Access Control. 2018 IEEE Conference on Communications and Network Security (CNS), Beijing, China, 2018, pp. 1-9. https://doi.org/10.1109/CNS.2018.8433209 (arXiv:1808.04581)

[11 - Aarohi Londhe] Atutxa, A., Sanz, A., Salegi, E., González, G., Astorga, J., & Jacob, E. (2026). IPsec based on Quantum Key Distribution: Adapting non-3GPP access to 5G Networks to the Quantum Era. arXiv preprint arXiv:2603.24426. https://arxiv.org/abs/2603.24426

[12 - Aarohi Londhe] Herzinger, D., Heise, L., Loebenberger, D., & Söllner, M. (2025). Time-Based State-Management of Hash-Based Signature CAs for VPN-Authentication. arXiv preprint arXiv:2509.11695. https://arxiv.org/abs/2509.11695

[13 - Aarohi Londhe] Moghimi, D., Sunar, B., Eisenbarth, T., & Heninger, N. (2019). TPM-FAIL: TPM meets Timing and Lattice Attacks. arXiv preprint arXiv:1911.05673. https://arxiv.org/abs/1911.05673

[14 - Aarohi Londhe] Perera, M., Mackay, M., Eiza, M. H., Raschellà, A., Shone, N., & Maheshwari, M. K. (2026). Towards Quantum-Safe O-RAN -- Experimental Evaluation of ML-KEM-Based IPsec on the E2 Interface. arXiv preprint arXiv:2601.20378. https://arxiv.org/abs/2601.20378

[15 - Aarohi Londhe] Gunasekara, R., Masood, R., & Kanhere, S. (2026). GETA: Generalized Encrypted Traffic Analysis. arXiv preprint arXiv:2605.31277. https://arxiv.org/abs/2605.31277

[16 - Aarohi Londhe] Pulls, T., & Witwer, E. (2023). Maybenot: A Framework for Traffic Analysis Defenses. arXiv preprint arXiv:2304.09510. https://arxiv.org/abs/2304.09510

[17 - Aarohi Londhe] Chehade, A., Ragusa, E., Gastaldo, P., & Zunino, R. (2025). Energy-Efficient Deep Learning for Traffic Classification on Microcontrollers. arXiv preprint arXiv:2506.10851. https://arxiv.org/abs/2506.10851

[18 - Pratik Shivekar] Chen, M., Chen, X., Ju, M., Zhao, D., & Gu, C. (2026). SeriCrypt: An LLM-Driven Context-Aware Serialization Framework for Cryptographic Protocols. arXiv preprint arXiv:2608.24498. https://arxiv.org/abs/2608.24498

[19 - Pratik Shivekar] Dvir, A., Zion, Y., Muehlstein, J., Pele, O., Hajaj, C., & Dubin, R. (2016). Robust Machine Learning for Encrypted Traffic Classification. arXiv preprint arXiv:1603.04865. https://arxiv.org/abs/1603.04865

[20 - Pratik Shivekar] Herzberg, A., & Shulman, H. (2009). Stealth-MITM DoS Attacks on Secure Channels. arXiv preprint arXiv:0910.3511. https://arxiv.org/abs/0910.3511

[21 - Pratik Shivekar] Annessi, R., Fabini, J., Iglesias, F., & Zseby, T. (2018). Encryption is Futile: Delay Attacks on High-Precision Clock Synchronization. arXiv preprint arXiv:1811.08569. https://arxiv.org/abs/1811.08569

[22 - Pratik Shivekar] Elliott, C., Pearson, D., & Troxel, G. (2003). Quantum Cryptography in Practice. arXiv preprint arXiv:quant-ph/0307049. https://arxiv.org/abs/quant-ph/0307049

[23 - Pratik Shivekar] Song-Ho, P., Myong-Suk, P., & Chung-Hyok, J. (2016). A method to Implement the Kerberos User Authentication and the secured Internet Service. arXiv preprint arXiv:1604.08799. https://arxiv.org/abs/1604.08799

[24 - Pratik Shivekar] Baseri, Y., Chouhan, V., & Hafid, A. (2024). Navigating Quantum Security Risks in Networked Environments: A Comprehensive Study of Quantum-Safe Network Protocols. Computers & Security, Volume 142, July 2024, 103883. https://doi.org/10.1016/j.cose.2024.103883 (arXiv:2404.08232)

[25 - Pratik Shivekar] Mink, A., Frankel, S., & Perlner, R. (2010). Quantum Key Distribution (QKD) and Commodity Security Protocols: Introduction and Integration. International Journal of Network Security & Its Applications 1.2 (2009) 101-112. https://arxiv.org/abs/1004.0605

Prepared by	**Issued by**	**Approved by**
Name: Parmeshwari Aland	Name:	Name:
Designation: Asst. Professor	Designation:	Designation:
Signature: P.V. Aland	Signature:	Signature: